

NATHAN TRAN

US Citizen · nathanantran.com · github.com/ntran306 · linkedin.com/in/ntran306
404-844-1962 · Jonesboro, GA 30236 · nathangsu306@gmail.com

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science

Minor in FinTech

Expected Fall 2027

GPA: 3.74

- Relevant Coursework: Machine Learning, Artificial Intelligence, Design & Analysis of Algorithms, Data Structures & Algorithms, Computer Organization & Programming, Database Systems, Systems and Networks, Linear Algebra, Finance & Investments

EXPERIENCE

Georgia-Pacific

Data Engineering Intern

May 2026 - August 2026

Atlanta, GA

- Developed an app catalog mapping and visualizing dependencies across 200+ applications to decrease troubleshooting time
- Provisioned AWS infrastructure with Terraform to automate dependency mapping and cost extraction across 12,000+ resources
- Delivered a certification workflow verifying app metadata, repos, resources, and dependencies, flagging 2,000+ optimization opportunities
- Identified \$100,000+ in potential annual cloud savings by surfacing untagged and unmapped resources

Autorobotics in Construction

Undergraduate Researcher

August 2025 - Present

Atlanta, GA

- Researching AI and VR applications in construction safety and education, building 6+ interactive Unity scenarios
- Prototyping instructional VR scenarios for an AI-guided training platform creating an adaptive learning environment
- Engineering cross-platform VR support via OpenXR for Meta Quest, SteamVR, and Oculus, sustaining 90+ FPS in PC-streamed builds

Viet Home Care LLC

Caregiver

January 2025 - Present

Gwinnett, GA

- Providing nursing support including diet monitoring, mobility assistance, and technological aid to improve client lifestyle
- Foster client well-being and independence through consistent emotional companionship
- Maintain client satisfaction and comfort through attentive care and reliable communication

PROJECTS

RatFinder (ByteFight Tournament)

Lead Developer

April 2026

Atlanta, GA

- Built a competitive game-playing AI agent that localizes a hidden target from noisy sensor data via a Bayesian belief filter
- Rebuilt a greedy baseline as an alpha-beta agent with transposition caching, lifting win rate vs. the benchmark bot 50%→80%+
- Tuned evaluation heuristics across 8+ agent versions, placing in the top 30% of 192 teams

FletchFlow

Solo Developer

July 2026 - Present

Atlanta, GA

- Created a computer-vision archery game using MediaPipe hand tracking to nock, draw, and release a virtual bow
- Designed 3-thread pipeline isolating 29 ms hand detection from the 60 FPS render loop, holding full rate on both
- Implemented One Euro filtering and a debouncing gesture state machine, stabilizing draw-and-release detection under hand jitter

Tutortle

Lead Full-Stack Engineer

August 2025 - March 2026

Atlanta, GA

- Led end-to-end development of location-based tutoring platform, improving tutor-student matching speed by 40%
- Increased session coordination by 25% through in-app messaging enabling faster connecting and scheduling
- Deployed 4+ APIs delivering real-time distance and travel-time estimates for matching

BuzzedIn

Full-Stack Developer

August 2025 - March 2026

Remote

- Shipped a Django job-matching platform connecting Georgia Tech students with recruiters
- Optimized database performance, simplifying calls and reducing load time to improve efficiency by 80%+
- Integrated 5+ APIs, boosting user engagement by 30% through dynamic filtering and responsive UX features

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C#, JavaScript, SQL, Lua, Assembly, HTML/CSS

Frameworks & Engines: Django, FastAPI, React | Unity, Unreal Engine

Libraries: MediaPipe, OpenCV, NumPy, pygame, moderngl, JAX, Flax, HuggingFace

Tools: AWS (EC2, S3, Batch, CloudWatch), Terraform, Docker, Git, GitHub, PostgreSQL, MySQL, PythonAnywhere, Twilio, Blender

Concepts: Full-Stack Development, Backend Development, Frontend Development, REST APIs, CI/CD, Automation, Agile/Scrum, DevOps

ADDITIONAL SKILLS & INTERESTS

Community Engagement: VGDev (GT Video Game Development Club), GT Swim Club, Volunteering

Interests: MMA, Fitness (calisthenics, bouldering, swimming), Piano, XR/VR applications, Video games

Languages: English (Native), Vietnamese (Elementary), Spanish (Elementary)